

Group Policy – Software Installation

IT Infrastructure Management (W04IST–SI0826G)

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1 Learning Objectives

By the end of this lab you will be able to:

1. Configure GPO-based software installation for both computers and users.
2. Distinguish between the *Assigned* and *Published* deployment modes and explain when each is appropriate.
3. Verify that Group Policy settings — including application installations, browser policies, and folder redirection — take effect on a client machine.
4. Deploy a second browser (Firefox) via GPO and enforce application-specific settings using its Administrative Templates.

2 Scenario

You are a junior sysadmin at Contoso, a company with regional offices in Sweden, Denmark, and Norway. Your manager has asked you to automate software deployment to workstations in the Swedish office, enforce a browser download policy domain-wide, redirect user Documents folders to a central file server, and lock down the guest account across all machines — all without visiting each workstation individually. You will use Group Policy to accomplish each of these tasks.

3 Before You Start

1. Open the ADUC console.
2. Create an Organizational Unit named **Sweden/Computers**.
3. Drag computer **CL1** from the **Computers** container to the OU **Sweden/Computers**.
4. Verify that the folder **C:\Install** is shared on the network.
5. Verify that the group **Authenticated users** has the **Read** permission on the **C:\Install** share.
6. Check whether the Chrome MSI installer is present in **C:\Install**.

7. Locate the **.admx** files for Chrome.
8. Copy all **.admx** files and the **en-US** subfolder to folder:

C:\Windows\SYSVOL\domain\Policies\PolicyDefinitions

If prompted to merge the **en-US** folder, confirm. This folder contains language-specific resources for the policy descriptions, and already exists. Merging it ensures that the Chrome policy nodes in the Group Policy editor show descriptions in English.

4 Exercise 1. Installing Software for a Computer

4.1 Why This Matters

When you assign software to a computer through Group Policy, it installs automatically at the next startup — no user interaction required. This is the standard way enterprise IT teams deploy tools to entire fleets of machines. Without this mechanism, you would need to visit every workstation individually or rely on users to install software themselves. Getting the deployment scope wrong (assigning to the wrong OU, or choosing the wrong path) silently prevents installation with no obvious error for the end user.

4.2 Task

1. On DC1, open the **Group Policy Management** console.
2. Open **contoso.internal** domain, and right-click **Group Policy Objects**. Select **New** and name it **AppForComputer**. Close the dialog.
3. Right-click **AppForComputer** and select **Edit**.
4. Navigate to **Computer Configuration / Policies / Software Settings / Software installation**.
5. Right-click the right pane and select **New > Package...**
6. In the file path field, enter the **network** path:

\\dc1.contoso.internal\install\Installers\GoogleChromeStandaloneEnterprise64.msi

Note: always use a UNC network path here, not a local drive path. The client machine must be able to reach this path over the network.

7. Click **Open**. Wait a moment — the package details are being read from the MSI.
8. In the **Deploy Software** dialog, select **Assigned** and click **OK**.
9. Close the GPO editor.
10. In the Group Policy Management console, right-click the OU **Sweden/Computers** and select **Link an Existing GPO...**
11. Select **AppForComputer** and click **OK**.

Before you restart CL1: will Chrome install on machines in all three country OUs (Sweden, Denmark, Norway), or only in Sweden? Give a reason based on where the GPO is linked.

1. Connect to the console of machine CL1.

2. Restart CL1 and watch the startup screen.

You should see a message “Installing managed software Chrome...” briefly appear before the logon screen.

3. Log in to CL1 as `contoso\administrator`.

4. Open the **Start menu** and verify that Chrome appears in the installed applications.

5. Log in to CL1 as a user from the **Norway** OU (e.g., `contoso\norsal1`).

6. Open the Start menu and check whether Chrome is installed for this user too.

Chrome should be installed for **all** users on CL1, regardless of their country OU — because the policy *Pis* is applied to the *computer*, not to the user.

7. On DC1, open the `AppForComputer` GPO editor.

8. Navigate to **Computer Configuration / Policies / Software Settings / Software installation**.

9. Right-click the `Chrome` entry and select **All Tasks > Remove...**

10. In the dialog that appears, select **Immediately uninstall the software from users and computers** and click **OK**.

11. Close the GPO editor.

After removing the package from the GPO, will Chrome uninstall immediately, or only after the next restart of CL1? What do you expect to happen?

1. Restart CL1.

2. Log in to CL1 as `contoso\administrator`.

3. Open the **Start menu** and confirm that Chrome is no longer listed.

If Chrome is still present, uninstall it manually via **Settings > Apps**.

4.3 Checkpoint

1. The GPO was linked to `Sweden/Computers`, but all users on CL1 could see the installed application regardless of their country OU. Why?

2. What is the difference between linking a GPO to a computer OU versus linking it to a user OU for software installation?

3. You chose a UNC network path (`\\dc1.contoso.internal\...`) instead of `C:\Install`. What would happen if you used the local path instead?

5 Exercise 2. Installing Software for a User

5.1 Why This Matters

Assigning software to a *user* instead of a computer shifts the installation trigger from startup to login. The Published mode goes a step further: the application is not installed automatically at all — it is advertised to the user, who can then install it on demand from Control Panel. This is useful for optional tools that not every user needs. The distinction between Assigned and Published determines whether your helpdesk gets calls about unexpected software appearing, or about users not being able to install what they need.

5.2 Task

1. On DC1, in the Group Policy Management console, right-click the OU **Sweden/Marketing** and select **Create a GPO in this domain, and Link it here...**
2. Name the new GPO **AppForUser** and click **OK**.
3. Right-click **AppForUser** and select **Edit**.
4. Navigate to **User Configuration / Policies / Software Settings / Software installation**.
5. Right-click the right pane and select **New > Package...**
6. Enter the network path to Chrome installer (the same as previously) and click **Open**.
7. In the **Deploy Software** dialog, select **Published** and click **OK**.
8. Close the GPO editor.

Before logging in: do you expect Chrome to be installed automatically when a Sweden/Marketing user logs on to CL1, or will they need to do something extra? What is the difference between Published and Assigned here?

1. On CL1, log in as a user from the OU **Sweden/Marketing** (e.g., **contoso\swemar1**).
2. Open the **Start menu** and check whether Chrome appears as an installed application.

Chrome should **not** be installed automatically — Published software is not pushed without user action.

3. Open **Control Panel**.
4. Select **Programs and Features**.
5. In the left panel, click **Install a program from the network**.

Chrome should appear in the list of available programs.

6. Select **Chrome** and click **Install**.
7. Confirm that the installation completes successfully.

5.3 Checkpoint

1. A *Published* application did not appear in the Start menu automatically. What must the user do to install it, and why is this sometimes preferable to Assigned?
2. Would a user in the **Norway/Marketing** OU see Chrome in “Install a program from the network”? Why or why not?
3. If you changed the deployment mode from Published to Assigned, what would change for the user at next logon?

6 Exercise 3. Applying Custom Application Settings

6.1 Why This Matters

Installing software is only the first step. Organisations routinely use Group Policy to enforce browser settings — download directories, homepage, proxy configuration — so that users cannot accidentally (or deliberately) change them. Administrative Templates (**.admx** files) translate application-specific registry settings into readable policy nodes in the Group Policy editor. Without them, an admin would have to configure each registry key manually. A misconfigured download path, for example, can send sensitive documents to an unmonitored location.

6.2 Task

1. On DC1, in the Group Policy Management console, right-click the **domain root** and select **Create a GPO in this domain, and Link it here...**
2. Name the GPO **ChromePolicy** and click **OK**.
3. Right-click **ChromePolicy** and select **Edit**.
4. Navigate to **Computer Configuration / Policies / Software Settings / Software installation**.
5. Right-click the right pane and select **New > Package...**
6. Enter the network path to the Chrome installer and click **Open**.
7. Select **Assigned** and click **OK**.
8. In the same GPO editor, navigate to **User Configuration / Policies / Administrative Templates / Google / Google Chrome**.

If the Google Chrome node is not visible, close the editor and return to the “Before you start” section to copy the **.admx** files, then re-open the editor.

9. Double-click **Set download directory**.
10. Select **Enabled**.
11. In the **Download directory** field, enter **C:\ChromeDownloads**.
12. Click **OK**.

13. Close the GPO editor.
14. Verify that the **ChromePolicy** GPO is linked at the **domain root** level (it should already be from step 1).
15. Restart machine CL1 and log in as any domain user.

Chrome should install automatically during startup.

16. Open Chrome and navigate to **Settings > Downloads**.

The download path should show **C:\ChromeDownloads** and the field should be greyed out (not editable).

17. In the Chrome address bar, type **chrome://policy** and press Enter.

The policy page should list **DownloadDirectory** with the value **C:\ChromeDownloads**.

18. Search for any small file on the Internet and download it.

19. Open File Explorer and navigate to **C:\ChromeDownloads**.

The downloaded file should be present in that folder.

The **C:\ChromeDownloads** path is a local directory used here for simplicity. In production, the download directory policy is typically set to a user-specific path (e.g., **\${user_home}\Downloads**) or a network share audited by the security team, not a shared local folder.

6.3 Checkpoint

1. Why were the Chrome browser settings greyed out in the Settings UI? What prevents the user from changing them?
2. The download directory policy was set under **User Configuration**. If you moved the same setting to **Computer Configuration**, would the behaviour change? Why?
3. What is the purpose of **.admx** files, and what happens if you try to configure Chrome policy without copying them to **PolicyDefinitions**?

7 Exercise 4. Advanced User Account Settings — Folder Redirection

7.1 Why This Matters

By default, a user's Documents folder lives inside their profile on the local machine. When a user logs on to a different computer, those documents are not available. Folder Redirection moves the physical storage of folders like Documents to a network share, while keeping the user experience (the path in File Explorer) unchanged. This also means the IT team can back up all documents centrally. If folder redirection breaks or is misconfigured, users may lose access to their files at logon — so getting the target path and permissions right is critical.

7.2 Task

1. On DC1, open **File Explorer** and create the folder **C:\docs** if it does not already exist.
2. Right-click the folder **C:\docs** and select **Properties**.
3. Open the **Sharing** tab and click **Advanced Sharing...**
4. Check **Share this folder** and set the share name to **docs**.
5. Click **Permissions**.
6. Select **Everyone** and remove all permissions.
7. Click **Add...** and add the group **Authenticated Users**.
8. Grant **Authenticated Users** the **Full Control** permission.
9. Click **OK** to close all dialogs.

Granting **Full Control** to **Authenticated Users** on a shared folder is a sandbox shortcut for simplicity. In production, use a dedicated security group (e.g., **G_Sweden_Marketing**) with **Change** permission, and reserve Full Control for administrators only. Apply the principle of least privilege.

1. In the Group Policy Management console, right-click the OU **Sweden/Marketing** and select **Create a GPO in this domain, and Link it here...**
2. Name the GPO **FolderRedirection** and click **OK**.
3. Right-click **FolderRedirection** and select **Edit**.
4. Navigate to **User Configuration / Policies / Windows Settings / Folder Redirection**.
5. Right-click **Documents** and select **Properties**.
6. In the **Setting** drop-down, select **Advanced – Specify locations for various user groups**.
7. Click **Add...**
8. In the **Security Group Membership** field, enter **G_Sweden_Marketing**.
9. Set the **Target folder location** to **Redirect to the following location**.
10. Enter the path: **\\dc1.contoso.internal\docs**
11. Click **OK** to close all dialogs.
12. Close the GPO editor.

Before logging in: where will the Documents folder physically be stored after this policy applies — on the local disk of CL1, or somewhere else? What will the user see in File Explorer?

1. On CL1, log in as a user from the OU **Sweden/Marketing** (e.g., **contoso\swemar1**).
2. Open **File Explorer**.
3. In the left panel, right-click the **Documents** folder and select **Properties**.
4. Click the **Location** tab.

The path should show **\\dc1.contoso.internal\docs\<username>** — confirming the folder is stored on the domain controller, not locally.

5. Log out and log in as **contoso\administrator**.
6. Open **File Explorer** and navigate to **C:\docs** on DC1.

You should see a subfolder named after the Sweden/Marketing user you just logged in as, containing their redirected Documents folder.

7.3 Checkpoint

1. The Documents folder appeared to be in the same place in File Explorer, but the Location tab showed a network path. Why does Windows show a consistent experience even though the storage location changed?
2. What would happen to a user's Documents if the domain controller became unreachable while they were logged on?
3. A user from the **Sweden/Sales** OU logged on to CL1 after this policy was applied. Was their Documents folder also redirected? Why or why not?

8 Exercise 5. Deploying and Configuring Firefox via GPO

8.1 Why This Matters

Supporting multiple browsers in an enterprise is common — different teams or applications may require Firefox instead of, or alongside, Chrome. The workflow mirrors what you did for Chrome, but Firefox ships its own set of Administrative Templates, which must be installed separately before any browser policies appear in the Group Policy editor. Knowing how to repeat this process for any application with ADMX support is a transferable skill: the same steps apply to Microsoft Edge, Adobe Reader, and many other managed applications.

8.2 Task

8.2.1 Preparation

1. On DC1, open a browser and navigate to the Mozilla Firefox enterprise download page:

<https://www.mozilla.org/en-US/firefox/enterprise/>

2. Download the correct version of the **Firefox installer** (64-bit) and save it to **C:\Install**.
3. On the same page, find and download the **Firefox Group Policy Templates** (a **.zip** archive).
4. Extract the archive. Inside you will find:
 - An **windows** subfolder containing **.admx** files and an **en-US** subfolder with **.adml** files.
5. Copy all **.admx** files and the **en-US** subfolder to:

C:\Windows\SYSTEM32\grouppolicy\policies\PolicyDefinitions

If prompted to merge the **en-US** folder, confirm.

8.2.2 Installing Firefox via GPO

1. Install Firefox using separate GPO, and assign it for computers in Sweden only.

The policy is linked to **Sweden/Computers**. A user from **Norway/Sales** logs on to CL1. Will Firefox be installed for that user? Justify your answer based on how computer-assigned software deployment works.

8.2.3 Configuring Firefox Settings via GPO

1. Right-click **FirefoxPolicy** and select **Edit**.
2. Navigate to **Computer Configuration / Policies / Administrative Templates / Mozilla / Firefox**.

If the Mozilla node is not visible, close the editor, re-check that the **.admx** files were copied correctly to **PolicyDefinitions**, and re-open the editor.

3. **Setting 1 — Disable private browsing:**
 1. Double-click **Don't allow Private Browsing**.
 2. Select **Enabled** and click **OK**.
4. **Setting 2 — Set the default homepage:**
 1. Double-click **Home page**.
 2. Select **Enabled**.
 3. In the **URL** field, enter **https://wit.pwr.edu.pl/**.
 4. Select **Locked** so users cannot change it.
 5. Click **OK**.
5. Close the GPO editor.

8.2.4 Verification

1. Restart CL1.

You should see a brief “Installing managed software Firefox...” message before the logon screen.
2. Log in to CL1 as **contoso\administrator**.
3. Open the **Start menu** and verify that Firefox appears in the installed applications.
4. Launch Firefox.
 - Confirm that the homepage is set to **https://wit.pwr.edu.pl/** and that the field is greyed out in **Settings > Home**.
5. In the Firefox address bar, type **about:policies** and press Enter.

The policy page should list both **DisablePrivateBrowsing** (set to **true**) and **Homepage** (with the URL you entered).
6. Attempt to open a **New Private Window** (Ctrl+Shift+P).

The option should be greyed out or absent from the menu, confirming the policy is enforced.

8.3 Checkpoint

1. Why must the **.admx** and **.adml** files be copied to **PolicyDefinitions** before Firefox settings appear in the Group Policy editor?
2. The homepage setting was configured under **Computer Configuration**. Would placing it under **User Configuration** change which machines or users it affects? Explain the difference.
3. A colleague suggests using a local path (**C:\Install\...**) instead of a UNC path when adding the Firefox MSI package. What would happen on CL1 at startup, and why?

9 Summary

In this lab you:

- **Configured** GPO-based software installation using both the Computer and User configuration nodes.
- **Distinguished** between the *Assigned* deployment mode (automatic, silent installation) and the *Published* mode (on-demand installation from Control Panel).
- **Verified** that policies applied correctly on CL1 — checking installed applications, browser download paths, folder redirection targets, and guest account settings.
- **Deployed** Firefox via GPO and enforced browser settings (homepage lock, private browsing restriction) using Mozilla's Administrative Templates.

These techniques form the foundation of enterprise workstation management. In real environments, GPOs are typically tested in a pilot OU before being linked domain-wide, and changes are tracked in a change management system to ensure rollbacks are possible.